Math 55 worksheet Monday 02/23/2026

1. Prove that a positive integer n is a perfect square if and only if every exponent in its prime factorization is even.

Solution: If all the exponents are even we can take the square root of each factor. Conversely, if $n = m^2$ then write $m = p_1 \cdots p_k$, so that $n = p_1^2 \cdots p_k^2$, and the claim follows from uniqueness in FTA.

2. Let p be a prime number, and let $\nu_p(n)$ denote the exponent of p in the prime factorization of n. If a and b are any two positive integers, prove that

$$\nu_p(a+b) \geq \min(\nu_p(a), \nu_p(b)).$$

Furthermore, if $\nu_p(a) \neq \nu_p(b)$, prove that

$$\nu_p(a+b) = \min(\nu_p(a), \nu_p(b)).$$

Solution: For the first statement, note that $p^{\min(\nu_p(a),\nu_p(b))}$ divides both a and b. For the other statement, suppose wlog that $\nu_p(a) < \nu_p(b)$. Let $r = a/p^{\nu_p(a)}$ and $s = b/p^{\nu_p(a)}$. Note that r and s are both integers, r is not divisible by p, but s is divisible by p. Then $p \nmid r+s$, so if we write $a+b = p^{\nu_p(a)}(r+s)$ it follows from uniqueness in FTA that $\nu_p(a+b) = \nu_p(a)$.

3. Let a and b be positive integers with prime factorizations $a = p_1^{e_1} \cdots p_k^{e_k}$ and $b = p_1^{f_1} \cdots p_k^{f_k}$. Show that
(a)
$$\gcd(a,b) = p_1^{\min(e_1,f_1)} \cdots p_k^{\min(e_k,f_k)},$$
(b)
$$\operatorname{lcm}(a,b) = p_1^{\max(e_1,f_1)} \cdots p_k^{\max(e_k,f_k)},$$
and deduce (c):
$$ab = \gcd(a,b)\operatorname{lcm}(a,b)$$

Solution: (a) Let $d = p_1^{\min(e_1,f_1)} \cdots p_k^{\min(e_k,f_k)}$. First show that $d \mid a, b$—this follows from $\min(e_i, f_i) \leq e_i, f_i$ for each p_i. Next, if $c = p_1^{g_1} \cdots p_k^{g_k}$ is any common divisor of a and b, then uniqueness in FTA shows that $g_i \leq e_i, f_i$ for all i, so $c \mid d$. Hence d is the gcd.
(b) is a completely parallel argument.
(c) follows from $e_i + f_i = \min(e_i, f_i) + \max(e_i, f_i)$.